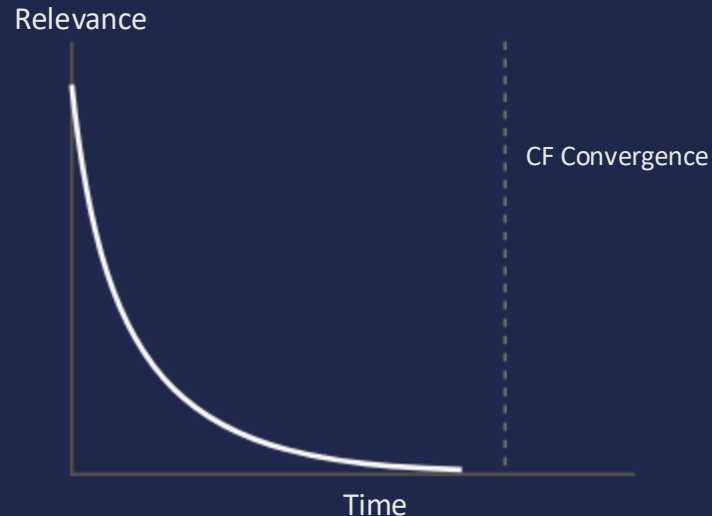


ROBUST ONLINE PERSONALIZATION

Kairos: Numerically Robust News Recommendation under Item Cold-Start via Cholesky-based LinUCB

The Item Cold-Start Mismatch: Why Static Recommenders Fail News Streaming

Temporal Mismatch



Articles become irrelevant before collaborative signals accumulate.

Representation Gap

User-Item Matrix

$$\begin{bmatrix} 0, & 0, & 0, & \dots \end{bmatrix}^T$$

→ Cold Start

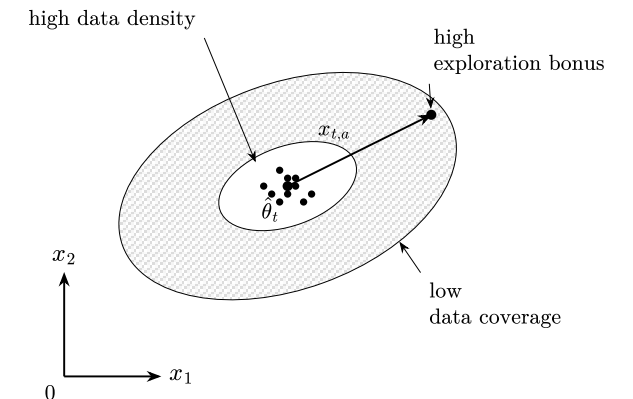
Text Encoder (MRL)

$$x = [0.82, -0.14, \dots]^T$$

→ Semantic Prior

Dense text embeddings eliminate the interaction vacuum.

The Bandit Engine



Uncertainty $\sqrt{x^T A^{-1} x}$
acts as an exploration bonus.

Online Exploration: LinUCB and the Limits of Recursive Inversion



Theoretical Formulation

- Payoff: $p_{t,a} = \hat{\theta}_t^\top x_a + \alpha \sqrt{x_a^\top A_t^{-1} x_a}$
- Design Matrix: $A_t = \sum_{\tau} x_{\tau} x_{\tau}^\top + I_d$
- Symmetric positive-definite (SPD) by definition
- Uncertainty represents Mahalanobis distance

LinUCB requires strict positive definiteness for valid exploration.

Numerical Reality

$$A_{t+1}^{-1} = A_t^{-1} - \frac{A_t^{-1} x x^\top A_t^{-1}}{1 + x^\top A_t^{-1} x}$$

- Sherman-Morrison rank-1 shortcut: $O(d^2)$
- Recursive outer-product subtractions accumulate drift
- Monte-Carlo (n=1000): Condition number $\kappa(A)$ diverges

Recursive subtraction breaks numerical stability over streaming horizons.

Square-Root Filtering: Guaranteed Stability via Cholesky Factor Updates

Factorization

Track precision factor directly: $A_t = L_t L_t^\top$

L_t is lower triangular with: $L_{ii} > 0$

Eliminates explicit inversion of A_t

Track square-root factors; avoid full matrix inversions.

Rank-1 Update

Givens Rotations: $[L_t \ x] \rightarrow [L_{t+1} \ 0]$

Maintains triangularity in $O(d^2)$ complexity

Additive orthogonal transformations (Gill et al. 1974)

Orthogonal rotations match Sherman-Morrison speed.

Inversion-Free Scoring

Forward substitution: $L_t z = x_a$

Quadratic form simplifies to:

$$x_a^\top A_t^{-1} x_a = \|z\|_2^2$$

Sum of squares guarantees unconditionally

$$\|z\|_2^2 = \sum_{i=1}^d z_i^2 \geq 0$$

Forward substitution guarantees strictly non-negative variance.

Latency Optimization: Subspace Search via Matryoshka Embeddings

Hierarchical Representations

- Nested training loss forces maximum semantic density into prefix slices
- Two-stage inference pipeline:
 - Stage 1 (Shortlisting): Fast candidate pre-filtering in $\mathbb{R}^{128}$ (MIPS)
 - Stage 2 (Re-ranking): Exact LinUCB exploration & scoring on Top-K in $\mathbb{R}^{768}$
- Eliminates the trade-off between semantic depth and streaming latency

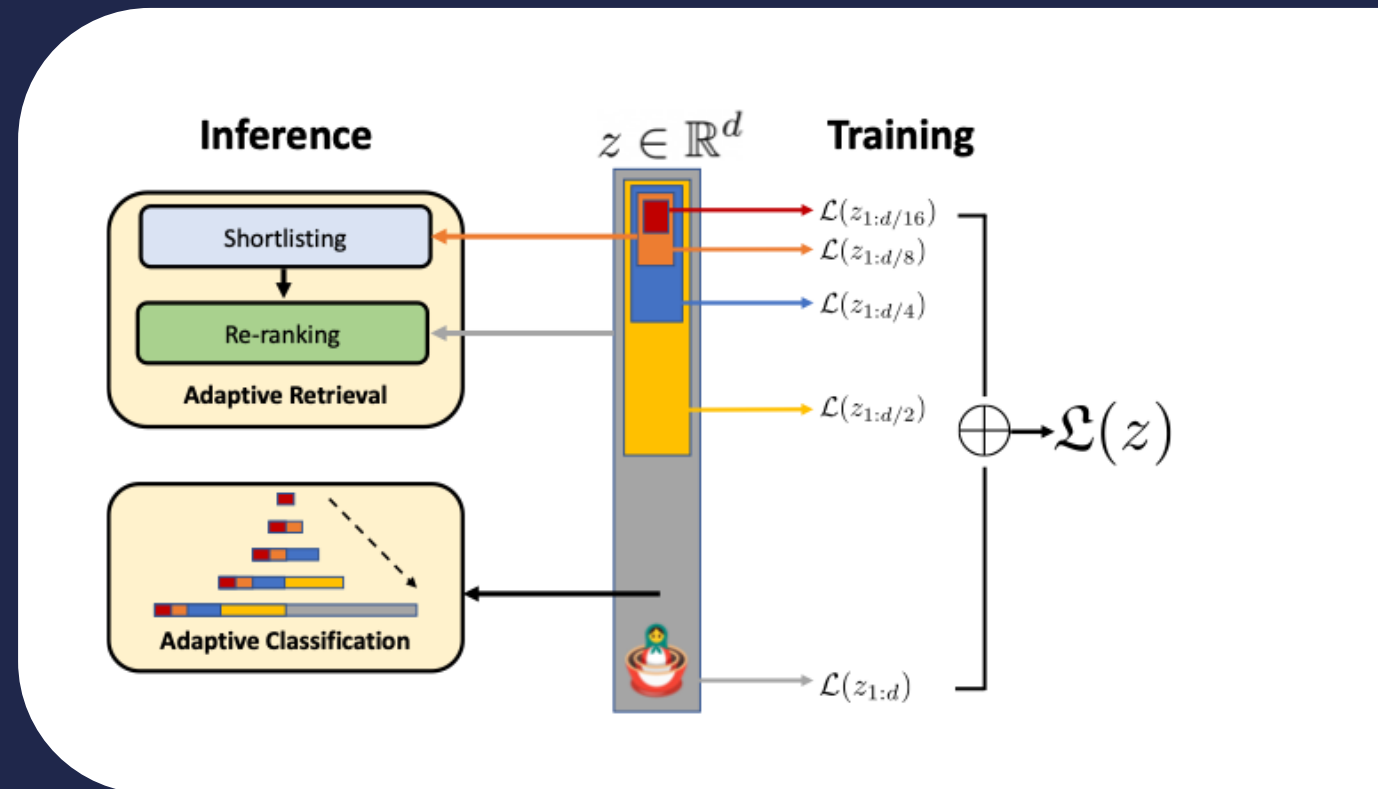


Fig. 1: Kusupati et al., *Matryoshka Representation Learning*, NeurIPS 2022

Empirical Evaluation: Tagesschau-API Streaming Benchmark

Stability & Condition Number

- 1,000 Monte Carlo runs over 1,000 streaming interactions
- Sherman-Morrison: Condition number κ diverges to $> 1,300$ with expanding variance
- Kairos (Cholesky): Condition number remains bounded ($\kappa \approx 1$) with near-zero variance

Cholesky preserves numerical conditioning under streaming collinearity.

Performance & Robustness

- Inference latency per 100 articles: 0.337 ms (768d) $\rightarrow$ 0.069 ms (128d)
- 4.85 $\times$ speedup (79.4% compute reduction) with MAE = 0.036 ± 0.022

MRL subspace retrieval delivers a 4.85 $\times$ speedup while preserving 96.4% structural fidelity.

Conclusion: Numerically Robust Online Personalization

Paper Contributions

- **Implicit SPD Guarantee:**
 - Cholesky factor updates eliminate recursive subtractions, preventing condition number divergence under streaming news features.
- **Adaptive Inference via MRL:**
 - 128d subspace retrieval achieves a 4.85× speedup (MAE = 0.036) without compromising ranking precision.
- **Cold-Start Mitigation:**
 - Directed uncertainty exploration enables immediate item visibility within narrow 48 h time-to-live windows.

Reproducibility & Next Steps

- **Code & Paper (GitHub):**



- **Outlook & Future Work:**
 - Hardware acceleration of Cholesky kernels for massive streams
- **Up Next:**
Live Demonstration: 3D Covariance Ellipsoid Drift

THANK YOU VERY MUCH FOR YOUR ATTENTION

Questions & Discussion